Homework

1. Let X be a discrete random variable taking values in the finite set $\{1, 2, \ldots, K\}$ for some fixed integer K. X has pmf
$$p(k) = \mathbb{P}[X = k] > 0, \quad k = 1, \ldots, K$$
and hence cdf
$$F(x) = \sum_{k=1}^{\lfloor x \rfloor} p(k),$$
$\lfloor x \rfloor = \max\{j \text{ an integer} \mid j \leq x\}$.

   Show that there are NO $c_n > 0$, d_n such that $F(c_n x + d_n)^n \to H(x)$ for a non-degenerate cdf H. Here, by "non-degenerate" we mean that H does NOT take the form $H(x) = \mathbf{1}_{x \geq x_0}$ for some $x_0 \in \mathbb{R}$, as this cdf corresponds to the non-random constant x_0. Thus, the limiting statement $\lim_{n\to\infty} F(c_n x + d_n)^n = H(x)$ for some non-trivial cdf H does not apply to discrete random variables taking a finite number of values.

2. We consider GPD for $U(0, 1)$ and $N(0, 1)$ below.

   (a) Let $X \sim U(0, 1)$. Find a function $\beta(u)$ so that
$$\lim_{u \uparrow 1} \sup_{0 \leq x \leq 1-u} |F_u(x) - G_{-1,\beta(u)}(x)| = 0.$$

   (b) Let $X \sim N(0, 1)$. Show that for $\beta(u) = 1/u$ and $\delta > 0$ that
$$\lim_{u \uparrow \infty} \sup_{\delta \leq x < \infty} |F_u(x) - G_{0,\beta(u)}(x)| = 0.$$

   In fact, the above also holds taking the maximum over $[0, \infty)$, rather than $[\delta, \infty)$, but this is very technical to show. **Hint:** To answer (b) above you may use without proof that for $x > 0$:
$$\frac{x}{1 + x^2} e^{-x^2/2} \leq \int_x^\infty e^{-y^2/2}\, dy \leq \frac{1}{x} e^{-x^2/2}.$$